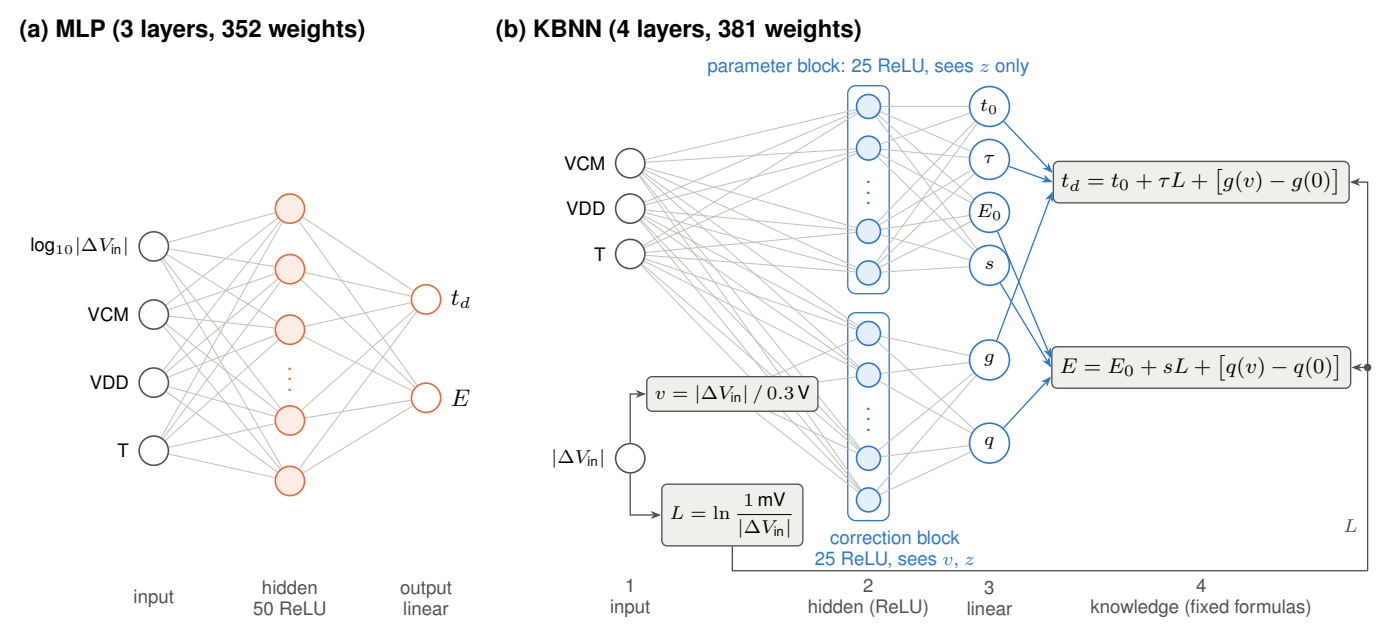

(a) MLP (3 layers, 352 weights)
(b) KBNN (4 layers, 381 weights)
parameter block: 25 ReLU, sees z only
$\log_{10}|\Delta V_{\text{in}}|$
VCM
VDD
T
t_d
E
t_0
τ
E_0
s
g
q
$t_d = t_0 + \tau L + [g(v) - g(0)]$
$E = E_0 + sL + [q(v) - q(0)]$
$v = |\Delta V_{\text{in}}| \,/\, 0.3\,\text{V}$
$|\Delta V_{\text{in}}|$
$L = \ln \dfrac{1\,\text{mV}}{|\Delta V_{\text{in}}|}$
L
correction block
25 ReLU, sees v, z
input
hidden
50 ReLU
output
linear
1
input
2
hidden (ReLU)
3
linear
4
knowledge (fixed formulas)